

Topic 4: Inorganic Chemistry and the Periodic Table

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include reacting Group 2 elements with water, heating nitrates and carbonates of Group 1 and 2 elements, investigating flame colours of s-block elements, preparing iodine from seaweed, investigating displacement reactions in the halogens, reacting Group 1 halides with concentrated sulfuric acid.

Mathematical skills that could be developed in this topic include manipulating data on the solubility of hydroxides.

Within this topic, students can consider how data can be used to make predictions based on patterns and relationships, for example by predicting properties of Group 7 elements.

Students should:

Topic 4A: The elements of Groups 1 and 2

1. understand reasons for the trend in ionisation energy down Group 2
2. understand reasons for the trend in reactivity of the Group 2 elements down the group
3. know the reactions of the elements Mg to Ba in Group 2 with oxygen, chlorine and water
4. know the reactions of the oxides of Group 2 elements with water and dilute acid, and their hydroxides with dilute acid
5. know the trends in solubility of the hydroxides and sulfates of Group 2 elements
6. understand reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved
7. understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions
Students will be expected to know the flame colours for Groups 1 and 2 compounds.
8. understand experimental procedures to show:
 - i patterns in thermal decomposition of Group 1 and 2 nitrates and carbonates
 - ii flame colours in compounds of Group 1 and 2 elements